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14 June 1977
SUPERSEDING
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MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL, POLYSTYRENE, GLASS FIBER REINFORCED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of glass fiber reinforced polystyrene suitable for injection molding and extrusion.

1.2 Classification. Glass fiber reinforced polystyrene molding material shall be of the following types, as specified (see 6.2).

Type I - High deflection temperature
Type II - High impact

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

FSC 9330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 256 - Impact Resistance of Plastics and Electrical Insulating Materials
D 570 - Water Absorption of Plastics
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 621 - Deformation of Plastics Under Load
D 638 - Tensile Properties of Plastics
D 648 - Deflection Temperature of Plastics Under Flexural Load
D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials
D 792 - Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

National Motor Freight Traffic Associations, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington DC 20036)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

3. REQUIREMENTS

3.1 Material. The molding or extruding material shall consist of suitably treated glass fibers dispersed in a polystyrene resin, with or without copolymers, elastomeric additives, colorants, and stabilizers, as required to meet the property requirements. There shall be no filler material, and only glass fibers shall be used as a reinforcing agent. There shall be no blending of unreinforced polystyrene with glass fibers. Uniform blending of unreinforced polystyrene with glass concentrates shall be permitted.

3.1.1 Glass fiber content. Total glass fiber content in the molding or extrusion material shall be between 34 and 38 percent inclusive.

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3.2 Property values. The material shall conform to the property values as specified in table I, when tested as specified in the applicable procedure of 4.3.

Table I. Property values

Property	Value required		Paragraph reference to applicable test
	Type I	Type II	
Tensile strength, pounds per square inch (psi), minimum (min.)	14,500	12,000	4.3.3
Izod impact strength 1/, foot pounds per inch of notch, min.	2.1	4.0	4.3.4
Deflection temperature under load at 264 psi fiber stress (unannealed), min., °C	104	88	4.3.5
°F	220	190	
Deformation under load (50° ± 1°C, 4000 ± 40 psi), percent maximum (max.)	0.25	1.5	4.3.6
Rockwell hardness, M scale, points, min.	93	85	4.3.7
max.	103	95	
Specific gravity, min.	1.33	1.31	4.3.8
max.	1.37	1.36	
Water absorption, weight gain plus soluble matter loss, percent, max.	0.13	0.20	4.3.9

1/ Izod impact strength is included for material identification only and does not predict nor correlate with performance of parts molded of these materials. Other impact tests, including tensile impact and drop weight impact, may show different relationships between impact behavior of these materials.

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3.3 Color. The color of the molding or extrusion material, including tolerances, and the method of determining compliance shall be as specified by the procuring activity (see 6.2). If color of molded plaques is required, the color and method of determination shall be as specified by the procuring activity (see 6.2)

3.4 Form. Unless otherwise specified, the form of the material (particle size and shape, and whether pelletized or unpelletized) and the method determining compliance, shall be as specified by the procuring activity (see 6.2).

3.5 Workmanship. The material shall be clean and contain no foreign material. The material shall be uniform in color, form, and all other appearance as determined visually. The uniformity shall be within each container and from container to container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for examination shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification will govern. A supplier's certificate of compliance with all the requirements specified in 3.1 and 3.1.1 shall be furnished.

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4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table II shall be approximately one pound.

Table II. Examination for defects in color, form, uniformity and appearance

Examine	Defect
Appearance and workmanship	Color 1/ not as specified
	Color not uniform
	Form 1/ not as specified
	Form not uniform
	Not uniform in other appearance
	Not clean, presence of foreign material

1/ When specified by the procuring activity, (see 3.3 and 3.4).

4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with table III, to determine that packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table III. Examination of preparation for delivery

Examine	Defects
Packing	Not level specified; not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability.
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location or sequence or method of application. Not in accordance with contract requirements

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

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4.2.3 Classification of tests. All tests shall be classified as follows:

(a) Lot acceptance tests (see 4.2.3.1).
(b) Periodic lot check tests (see 4.2.3.2).

4.2.3.1 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot except when periodic lot check tests are required. The lot acceptance tests shall be as specified in table IV, for types I and II material. Unless otherwise specified by the procuring activity, lot acceptance results shall be included with each lot supplied.

Table IV. Lot acceptance tests

Test
Tensile strength
Izod impact strength (for type II only)
Deflection temperature under load (for type I only)
Rockwell hardness
Specific gravity

4.2.3.2 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring activity (see 6.2). Periodic lot check tests shall consist of all the tests specified in table I for types I and II material. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot. Unless otherwise specified by the procuring activity, periodic lot check test results shall be included with each lot requiring periodic lot check tests.

4.2.4 Testing. The material shall be tested for the applicable characteristics listed in table I, in accordance with the test methods specified, for each lot submitted for inspection. The sample unit shall consist of sufficient material to prepare all specimens required for testing. The test result recorded for each characteristic of each sample unit shall be the averaged results from the specimens for that unit. The number of sample units selected from each lot shall be in accordance with the following:

Lot size (pounds)	Number of sample units (with no failures)
Up to 10,000	2
10,000 to 100,000	3
Over 100,000	5

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4.3 Test methods.

4.3.1 Specimen preparation. Test specimens shall be prepared by injection molding in accordance with recommendations of the manufacturer.

4.3.2 Specimen conditioning and testing. Test specimens shall be conditioned in accordance with procedure A of ASTM D 618 and tested at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity.

4.3.3 Tensile strength and elongation. Three specimens shall be tested in accordance with ASTM D 638, using testing speed B. Dimensions of each specimen shall conform to type I of figure 1 of ASTM D 638.

4.3.4 Impact strength. Three specimens shall be tested in accordance with ASTM D 256, using method A. Specimens shall be 0.25 by 0.5 inch bar.

4.3.5 Deflection temperature. Three specimens shall be tested in accordance with ASTM D 648. Specimen thickness shall be 0.25 inch.

4.3.6 Deformation under load. Three specimens shall be tested in accordance with ASTM D 621. Each specimen shall be a 1/2 inch cube cut from a 1/2 by 1/2 inch bar and tested with the load applied in a direction parallel to the flow of material in the original bar.

4.3.7 Rockwell hardness. A minimum of five readings shall be taken equally spaced along the length of three 1/2 by 1/2 by 5 inch test bars in accordance with ASTM D 785.

4.3.8 Specific gravity. One determination shall be made in accordance with ASTM D 792. Method A1 or A2 of ASTM D 792 shall be used, with the specimen conforming to dimensional requirements of ASTM D 792.

4.3.9 Water absorption. Three specimens shall be tested in accordance with ASTM D 570, using the 24 hour immersion period.

5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

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5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds, maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Freight Classification Rules, as applicable to the mode of transportation employed.

5.2 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Glass fiber reinforced polystyrene is used for applications that do not justify the cost of laminating but do require a material possessing high strength and stiffness, resistance to creep and corrosion, and nonmagnetic properties. Type I materials provide dimensional stability at elevated temperatures, while type II materials exhibit superior resistance to shock and impact. Glass fiber reinforced materials are suitable for fabricating various components of nonmagnetic land mines, switch parts, fuses and other items.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of material required (see 1.2).
(c) Color, if required, including use of molded plaques, if required (see 3.3).
(d) Form, if required (see 3.4).
(e) Periodic lot check tests when required (see 4.2.3.2).
(f) Level of packing required (see 5.1).

6.3 Formulation. Polystyrene materials that meet the requirements of this specification have been formulated with glass contents of 34 to 38 percent in the final item. Type I materials generally consist of unmodified polystyrene, while type II materials are formulated with small quantities of copolymers of elastomeric additives to enhance impact resistance.

Custodians:

Army - MR
Navy - AS
Air Force - 11

Review activities:

Army - AR, EL, ME
Air Force - 11, 99
DLA - GS

User activities:

Army - AT
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Preparing activity:
Army - MR

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